

Education

University of Wisconsin-Madison

B.S. Mathematics, Statistics and Computer Science Certificate

2024- GPA: 3.95

Relevant Coursework: Linear optimization, Data Science Modeling, Applied Regression Analysis, Data Structure, Probabilistic Graphical Model, Probability Theory, Mathematical Statistics, Linear Algebra, Mathematical Analysis, Machine Learning.

Experience

Multi-Factor Quantitative Trading Strategy Development

February 2025

- Designed and implemented a stock selection system based on multi-factor models, analyzing S&P 500 constituent stocks data using Python
- Built a comprehensive factor library including 10+ factors across momentum, value, quality, and volatility categories, and evaluated factor effectiveness using IC/IR metric
- Applied machine learning methods (Random Forest) to optimize factor weights and constructed a dynamic asset allocation model
- Technologies: Python, pandas, numpy, scikit-learn, matplotlib, Backtrader

Experience

MadLuv - Campus Social Connection Platform

February 2025

- Developed a gradient boosting model for the partner matching system using speed dating dataset from Kaggle
- Performed comprehensive data cleaning and feature engineering to improve prediction accuracy
- Designed and implemented a sophisticated data pipeline for processing user profiles into the model's required format
- Led the overall application architecture design, integrating matching algorithms with activity recommendation systems
- Created a community-focused platform that addresses dual challenges of campus connections and local activity discovery

Sea Lamprey Population Dynamics Modeling

Modeling and Programming Lead

February 2024

- Developed and analyzed ordinary differential equation models to study the impact of sea lamprey sex ratios on the ecological system
- Created a logistic growth model to explain the natural population growth of both male and female sea lampreys, incorporating various assumptions. Designed and tested a series of models (predator-prey, competitive species) to simulate different ecological scenarios.
- Evaluated the effects of metamorphosis and sex ratio changes on population equilibrium using parameter sensitivity analysis.

Relation Evolution Analysis Based on Dynamic Knowledge Graph and Dynamic Bayesian Networks

March 2025 - present

- Designed and implemented a relation evolution analysis framework combining Dynamic Knowledge Graphs (DKGs) and Dynamic Bayesian Networks (DBNs) to capture temporal patterns in entity relationships
- Extracted entities and relationships from a comprehensive dataset of news headlines spanning 52 media outlets (2017-2021)
- Applied advanced NER and relation extraction techniques using LLMs t

- Constructed time-sliced knowledge graphs to model relationship evolution over time, utilizing Neo4j and NetworkX
- Implemented probabilistic inference using DBNs (pgmpy) to model dependencies between entities and predict future relationship developments

Skills

Skills: Python, R, Scikit-learn, LaTeX, pytorch